Thm Projective spaces $\mathbb{P}^n$ are D-affine $= \begin{cases} \Gamma \text{ exact} \\ \Gamma \text{ conservative.} \end{cases}$ $\overset{\text{Bern-Bell}}{\Longrightarrow}$ $\Gamma : DMod(X) \xrightarrow{\simeq} Mod(\Gamma(X))$

Pf (1) exactness: $V^\times = V - \{0\}$ NOT affine

$V^\times \downarrow p \; U$

$$\underbrace{\mathbb{P}(V) = V^\times/\mathbb{G}_m}_{\text{equivariant, exact}} \xhookrightarrow{j} V/\mathbb{G}_m \xrightarrow{\text{push}} \mathbb{C}/\mathbb{G}_m \xrightarrow{\text{push}} \mathbb{C}.$$

sh: rational $\mathbb{G}_m$-rep $=$ graded vector sp (weight)

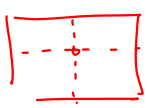
$\mathbb{C}/\mathbb{G}_m$: forgetful $\mathcal{O}(U)$-str.

push: $\mathbb{G}_m$-inv $=$ weight 0

exact

$\mathcal{O}(m) \longmapsto \bigoplus_{k\geq 0} \Gamma(\mathbb{P}^n, \mathcal{O}(m+k))$ weight k $\longmapsto$?? cuts where cohomology $\xrightarrow{m\gg 0} \Gamma(\mathbb{P}^n, \mathcal{O}(m))$

$V^\times$



Take ses. of D-modules $0 \to \mathcal{M}_1 \to \mathcal{M}_2 \to \mathcal{M}_3 \to 0$ in $DMod(\mathbb{P}^n)$

Apply functor $0 \to j_\bullet p^+ \mathcal{M}_1 \to j_\bullet p^+ \mathcal{M}_2 \to j_\bullet p^+ \mathcal{M}_3 \to 0$ in $DMod^{\mathbb{G}_m}(V)$

$\overset{\shortparallel}{j_\bullet p^* \mathcal{M}_1}$ etc ...

$j_\bullet$ is not exact, might have cohomology.

$z \in \mathbb{G}_m$

$z \cdot v = z^k v$

$\frac{d}{dz}$

Pb not of functors are exact, to show Γ is exact $\Longleftrightarrow$ cohomology is not in weight 0.

Observe: detect weight using Euler vector field (generates $\mathbb{G}_m$-action):

Exercise: $\Theta = \sum x_i \partial_i$ act on weight k piece by eigenvalue k.

Observe: cohomologies $H^\bullet$ must supported at $0 \in V$

Koszul: $i : \{0\} \to V$ $\quad i_! N = i_! \mathbb{C}_k = \bigoplus i_! k$ ← Check the claim for this

Compute: $i_! k = D_V / D_V \langle v \partial \rangle \simeq \mathrm{Sym}\, T_V$ $\quad$ Claim $\Theta \cdot \mathrm{Sym}^d T_V$ by $-(d+1)n$

Check: $d=0$: $\underset{1}{\Theta \cdot 1} = \sum x_i \partial_i \cdot 1 = \sum (-1 + \partial_i x_i) \cdot 1 = \sum -1 \cdot 1 = -n \cdot 1$ ✓

② conservative, i.e. if $M \neq 0$ then $\Gamma(M) \neq 0$.

Say $M \neq 0$, need j_*j^*M has weight 0 part. (on $V/\mathbb{G}_m$)

- $j_*j^*M \neq 0$ b/c restrict to year...
- Say it has a positive weight. Use ∂_i to lower weights.
  Say $\partial_i m = 0$ $\forall i$ $\Rightarrow$ weight$(m) = 0$. $\theta = \sum x_i \partial_i \cdot m = 0$
  So, get sth in wt 0.
- Say it has negative weights. If x_i don't all act by torsion, get wt 0. Otherwise, j_*j^*M supp. at 0. (Kashiwara) $\Rightarrow\Leftarrow$

$\Rightarrow$ Has weight 0 part $\Rightarrow \Gamma(M) \neq 0$. $\square$

Remark: simpler

$$\begin{tikzcd} G/U \arrow[r] \arrow[d, "U\text{-torsor}"] & \mathrm{Aff}(G/U) \arrow[r, hook, "\Phi"] & \prod V_i \arrow[d] \\ G/B \arrow[rr] & & \prod \mathbb{P}(V) \end{tikzcd}$$

Remark: What X are D-affine?

- G/P partial flag var.
- affine X

Conjecture: these are the only projective cases

Thm (Beilinson-Bernstein) $\mathfrak{Z}_{\mathfrak{g}} \subseteq \mathcal{U}_{\mathfrak{g}}$ (center $\mathfrak{Z}_{\mathfrak{g}}$, univ. env. alg. $\mathcal{U}_{\mathfrak{g}}$) k_0 = augm. char. of $\mathfrak{Z}_{\mathfrak{g}}$ (module)

G acts on G/B $\rightsquigarrow$ $\mathcal{U}_{\mathfrak{g}} \longrightarrow D(G/B)$

① factors thru equiv. $\mathcal{U}_{\mathfrak{g}} \otimes_{\mathfrak{Z}_{\mathfrak{g}}} k_0 \xrightarrow{\simeq} D(G/B)$

important means strong equiv. i.e. condition not extra structure.

② G/B is D-affine

strongly U-equiv. D-mod on G/B.

$\Rightarrow$ $\Gamma(G/B, -): \mathrm{DMod}(G/B) \xrightarrow{\simeq} \mathrm{Rep}(\mathfrak{g})_0$ (trivial central char.)

$\cup\!\!\mid$ $\cup\!\!\mid$

$\mathrm{DMod}(U\backslash G/B) \simeq \mathrm{DMod}_{\Lambda}(G/B) \xrightarrow{\simeq} \mathcal{O}_0^{geom}$ trivial block of "geom" cat $\mathcal{O}$

(diff. alg $\mathcal{O}$: $\mathfrak{h}$-semisimple
geom $\mathcal{O}$: $\mathfrak{Z}_{\mathfrak{g}}$-semisimple)

where Λ is s.s. condition

"conormals to U-orbits on G/B" $U \curvearrowright G/B$

X/G

weak eq	strong eq
M. D-mod eq sh.	M D-m.s. $\rightsquigarrow$ $\mathfrak{g} \curvearrowright D_X$ eq. ch. $\rightsquigarrow$ $\mathfrak{g} \curvearrowright \Omega M$
+ commutating	argue

Ex.: strong eq. $\mathbb{G}_m$ on $*$ $\longleftrightarrow$ trivial

derived cat "as usual"

$\mathrm{Hom}(\mathbb{G}_m, \mathbb{G}_m)$

weak: have meaning.

derived cat
- dg categories
- $\downarrow$ (redundant equations)

strong: $*$ trivial

$\mathbb{G}_m$

Part 1

$$\begin{tikzcd} U\mathfrak{g} \arrow[r, "\alpha"] \arrow[d, "gr"] & D(G/B) \\ \mathrm{Sym}\,\mathfrak{g} \arrow[r, "\mu"'] & \mathcal{O}(T^*G/B) \end{tikzcd}$$

$\ker(\alpha) \supseteq \ker(U\mathfrak{g} \to U\mathfrak{g} \otimes_{Z\mathfrak{g}} k)$

$\cap \qquad\qquad \subseteq$

has classical limit.

rational res of singularities etc...

$$\begin{tikzcd} \tilde{\mathcal{N}} \arrow[r, hook] \arrow[d] & \tilde{\mathfrak{g}} \arrow[d] \\ \mathcal{N} \arrow[r, hook] \arrow[d, "\Gamma"] & \mathfrak{g} \arrow[d, "\text{Chevalley restr. thm}"] \\ \{0\} \arrow[r, hook] & \mathfrak{g}/\!/G \end{tikzcd}$$

"lacks $\hbar$"

$U\mathfrak{g} \otimes_{Z\mathfrak{g}} k$

$\wr$

$\boxed{\mathcal{O}(T^*G/B)} \otimes k$

$\mathcal{O}(\mathfrak{g}/\!/G)$

H-C

$\simeq$

$\mathcal{O}(\mathcal{N})$